

1 Additive-input encoding fails target-held-out prediction on current Perturb-seq screens

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1.1 Abstract

Pooled Perturb-seq screens are commonly modelled with a linear settled-state assumption and an additive input per perturbed gene, under which the response operator is identifiable in closed form as $\mathbf{J} \cdot \mathbf{P} \cdot \mathbf{X} = -\mathbf{U} \cdot \mathbf{S}^+$. We tested whether it predicts held-out perturbations in three screens — Replogle K562 essential, Replogle RPE1 essential, and Jost 2020 — using target-held-out nested cross-validation against a matched-SNR linear-truth control. The model failed on all three: real held-out ρ stayed at the predict-zero baseline (K562 0.96 with 5-fold SD 0.02; RPE1 1.00, SD 0.006; Jost 1.00, SD 0.002) while the matched linear truth reached 0.18, 0.53, and 0.72. The failure survives random target sampling (K562 real 1.00 vs matched 0.43), reducing input dimension to an overdetermined regime (Jost $d = 5$ real 1.01 vs matched 0.32), and column-normalization of the design, so mis-estimated knockdown efficiency cannot explain it. It is not a noise problem: at the observed signal amplitude the same estimator recovers a known operator’s interactions (interaction-only cosine 0.98 [K562], 0.74 [RPE1], cross-replicate null ≈ 0). On Jost the pipeline partly interpolates dose along known target directions (guide-level $\rho = 0.66$ vs matched 0.22). A preregistered alternative encoding based on control-cell co-expression did not generally rescue prediction (it helped on one K562 target selection only), so whether the failure lies in the projection of targets onto programs or in nonlinear dose response remains open.

1.2 1. Introduction

Pooled Perturb-seq screens deliver an intervention→response map at genome scale by pairing CRISPR-based perturbations with single-cell RNA-seq [1]. A common modeling step reads these responses through a linear settled-state model of regulatory dynamics: knocking down gene g at efficiency κ_g shifts program-space state by $\Delta \mathbf{z}_g = -\mathbf{J}^{-1} \mathbf{u}_g$, where the perturbation input is $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \delta_g$, with $\mathbf{W}$ a control-derived program basis and δ_g a one-hot on gene g . Under this additive-input steady-state assumption, stacking guides gives a sensitivity matrix $\mathbf{S} = -\mathbf{J}^{-1} \mathbf{U}$, and regularized inversion returns the operator’s action on the identified response subspace, $\mathbf{J} \cdot \mathbf{P} \cdot \mathbf{X} = -\mathbf{U} \cdot \mathbf{S}^+$ with $\mathbf{X} = \text{range}(\mathbf{S})$. Because this action is closed-form, the pipeline that computes it can be run on any Perturb-seq screen with a defined program basis and a calibrated efficiency estimator.

Steady-state perturbation inference has a lineage in the systems-biology literature that predates Perturb-seq: modular response analysis and reverse-engineering methods reconstruct network structure from small numbers of steady-state perturbations under an assumed linear response [3–5]. Recent Perturb-seq-era methods extend that program to genome-scale interventional data with linear or linear-cyclic structural models in gene space, including instrumental-variable, DAG, and latent-linear approaches [6–9]. At genome scale, projecting responses onto a low-dimensional program basis is the natural reduction, and $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \delta_g$ is its direct program-space form. This paper tests whether that reduction preserves target-held-out predictability. A recent benchmark separately found that a simple linear model predicts unseen

single-gene perturbations in Replogle-type screens about as well as current deep-learning models [10] — a finding that sits alongside our result and helps place the failure at the encoding stage rather than the model-class stage (§3.2). Answering the primary question requires a fold-honest evaluation: fit $\mathbf{A} = -\mathbf{U}\cdot\mathbf{S}^+$ on training guides, predict $\mathbf{U}_{\text{test}} = -\mathbf{A}\cdot\mathbf{S}_{\text{test}}$ on held-out guides, and compare the resulting held-out ρ to a positive control that runs the same procedure on a linear ground truth simulated at the dataset’s own guide inputs, efficiency, and per-entry noise. Under any additive-input linear model the positive control must beat the predict-zero baseline of $\rho = 1$; otherwise the identifiability calculation is not informative on the data at hand.

We ran that control on three current Perturb-seq screens: the essential-gene libraries in Replogle et al. 2022 (K562 and RPE1 lines) [1] and the titrated dose-response library in Jost et al. 2020 [2]. On all three, the fitted encoding fails to beat the predict-zero baseline on held-out prediction, while the matched-SNR linear-truth control does. The failure is not a noise-level artifact: at the observed signal amplitude the same estimator on a linear ground truth cleanly recovers the interaction structure of the operator. It is not created by the widely used convention of selecting the 200 targets with the most cells: a random-200 K562 refit reproduces the failure. It is not created by identifiability under-determination on Jost’s smaller target set: a d-sweep truncating the coordinate system to a comfortably overdetermined regime leaves real held-out ρ at the predict-zero floor. And it is not created by mis-scaling of the perturbation-strength estimate: column-normalizing the design before the fit removes all per-guide scale that a κ error could carry, and held-out ρ under the same nested cross-validation stays at the predict-zero baseline on all three screens (0.97, 1.00, 0.99) while the matched linear truth reaches 0.43, 0.78, and 0.79.

Two candidate explanations remain live. Under the additive-input encoding, projecting the target-direction one-hot onto $d = 30$ control-derived programs leaves $\sim 99\%$ of the target direction outside the retained subspace, so the encoded input may carry little of the response-relevant signal — a *projection* failure. Independently, the linear settled-state assumption may be the wrong shape for saturating or threshold-dominated dose responses that govern actual regulatory dynamics — a *dose* failure. We describe the tests that would distinguish them and leave both to future work.

Two subsidiary correctness issues bear on any tool in this space and are handled in the software. Eps-based numerical rank on collinear guide libraries can silently declare full rank; the identifiability gate now refuses to declare full-domain recovery unless both the response and the input have full rank. And knockdown-efficiency estimators can be applied outside their valid data format; the efficiency router chooses `mean_ratio` on count data and a `detection_rate` proxy score on pre-scaled residual data, with the second labelled as a signed distributional shift rather than a fractional-knockdown estimate.

anchor-op implements the pipeline, the identifiability discipline, the efficiency-routing regime, the matched-geometry positive control, and the target-held-out nested cross-validation as a reusable set of diagnostics. All numeric claims in this manuscript are round-form values; the exact values, the JSON they were computed from, and the script that produced each are listed in a machine-readable provenance table (Code availability).

1.3 2. Results

Figure 1. (a) anchor-op fitting pipeline: program-space projection of expression data on a control-derived PCA basis, encoding of each perturbation input, and a regularised target-grouped-fold operator fit. (b) Matched-SNR linear-truth positive control used throughout: simulate a known linear ground truth at the dataset’s own $\mathbf{U}$, κ , and σ , then run the same

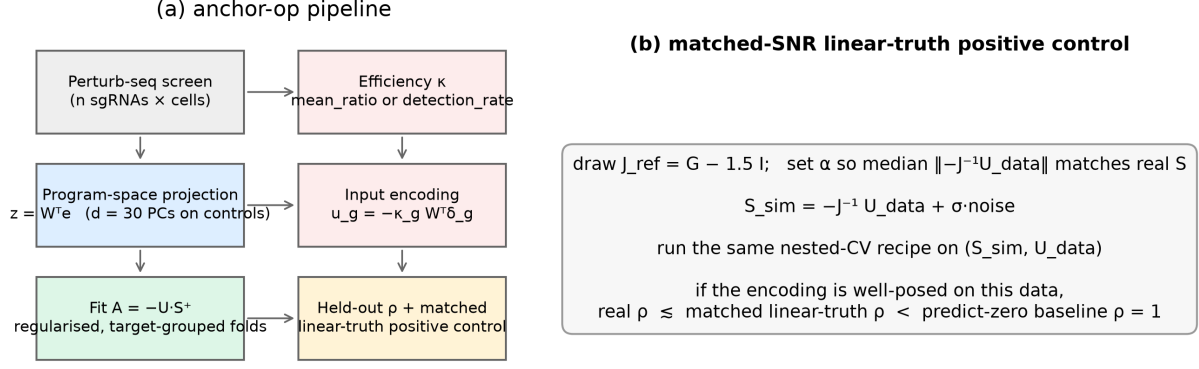


Figure 1: Figure 1

nested-CV recipe on the simulated (S, U) pair. If the encoding is well-posed on this data, real ρ should track the matched-linear-truth ρ below the predict-zero baseline of 1.

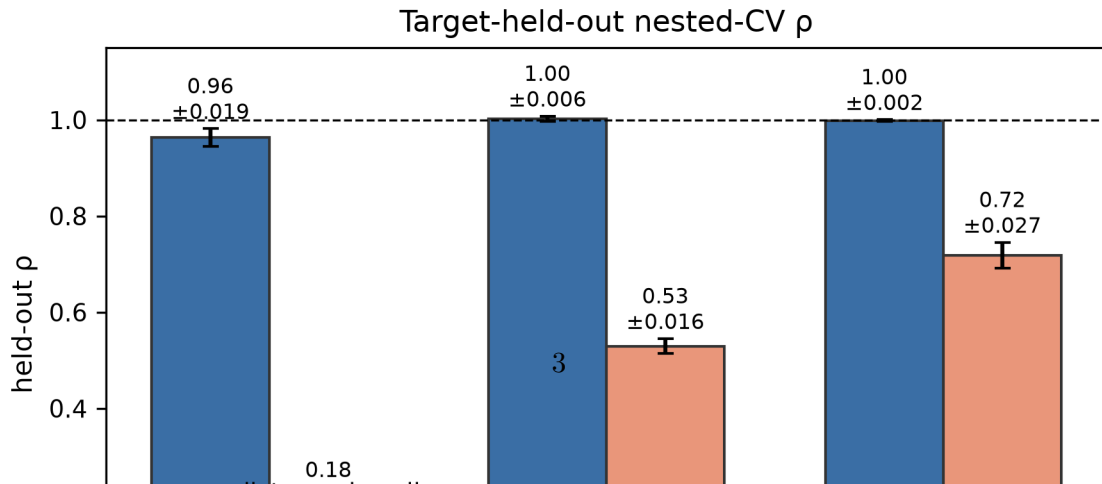
1.3.1 2.1 Target-held-out prediction fails against the matched linear control

For each screen we fit the additive-input operator $A = -U \cdot S^+$ using truncated SVD, with the retained rank chosen by nested cross-validation inside each outer training fold (Methods). Both outer and inner folds are target-grouped, so no sibling sgRNA leaks between train and test. The rank grid includes zero, which represents the predict-zero baseline $A \equiv 0$ and gives held-out $\rho = 1$ exactly; nested CV therefore selects “shrink to zero” whenever the fitted model does worse than predict-zero on inner validation.

We report pooled held-out ρ with its across-outer-fold standard deviation on the real data, and mean $\pm$ SD across 15 independent matched-SNR linear-truth simulations. Table 1 summarizes.

Table 1. Target-held-out ρ under nested cross-validation.

Screen (n retained sgRNAs / n targets)	Real ρ (5-fold SD)	Median picked TSVD rank	Matched linear-truth ρ (mean, SD; N = 15)
Replogle K562 essential (188 / 188)	0.96 (0.02)	3	0.18 (0.003)
Replogle RPE1 essential (153 / 153)	1.00 (0.006)	1	0.53 (0.016)
Jost 2020 (122 / 25)	1.00 (0.002)	0	0.72 (0.043)



linear-truth control, orange). Error bars: 5-fold SD for real, cross-replicate SD across 15 sims for the control. Dashed line: predict-zero baseline $\rho = 1$.

Every real ρ sits at or above the predict-zero baseline within one fold-level SD. On K562 the median picked rank is 3; on RPE1 it is 1; on Jost it is 0 — nested CV picks “shrink to zero” as the best per-fold model, so the fitted operator loses to no model at all. Under the same estimator on the same U and per-entry σ , a matched-SNR linear ground truth reaches held-out ρ well below the predict-zero baseline in every case.

The three screens differ in library size and design. Replogle-essential libraries retain one sgRNA per target after the released target-level aggregation step. Jost carries several sgRNAs per target at graded activities, spanning $\kappa \in [0.07, 1.00]$ with count-based estimates from `mean_ratio` on UMIs — roughly twice Replogle’s κ range in a design purpose-built for dose response — and 25 targets total. Their per-entry noise anchors, estimated from within-guide cell-level split-half bootstraps in each screen’s own basis, are 0.24 (K562), 0.35 (RPE1), and 0.066 (Jost, per sgRNA). The failure is not restricted to a specific library size, design, or noise level.

1.3.2 2.2 The failure is not a noise problem

At the amplitude of the released simulation ensemble the noise-free response is much smaller than the observed response — a factor of ~ 360 on K562 by median column norm. At that amplitude the signal-to-noise ratio is low and any TSVD estimator, linear-truth or not, has almost no chance of recovering the operator. A signal-scale sweep varying the multiplier α_S on the noise-free response confirms this and identifies the α_S that reproduces the observed median column norm of S .

At the data-matched amplitude the same TSVD estimator on a linear ground truth cleanly recovers the operator: full-matrix Frobenius cosine 0.99 (K562) and 0.91 (RPE1) at $N = 200$ replicates. Those full-matrix cosines are dominated by the shared diagonal shift carried by every replicate in the ensemble; their cross-replicate null is around 0.70. The *interaction-only* cosine — computed after subtracting the diagonal from both A and J — isolates the recoverable structure. It reaches 0.98 (SD 0.003) on K562 and 0.74 (SD 0.023) on RPE1 at matched amplitude, against a cross-replicate null of 0.003 in both cases ($N = 200$).

The estimator can recover a known operator’s interactions at the signal amplitude of the observed response. The gap between “an estimator that recovers a linear operator’s interactions at matched SNR” and “an estimator that fails held-out prediction on real data at the same SNR” is the point. Under the additive-input encoding, the model fits A and predicts $U = -A \cdot S$ on held-out guides. On the linear-truth positive control that prediction succeeds; on the real data it does not. The failure lies somewhere between the data-generating process and the additive-input model, not in the fitting machinery.

Figure 3. (a) Signal-scale α_S sweep of full-matrix Frobenius cosine on K562 and RPE1, with the shift-1 cross-replicate null on each curve. Vertical bars mark the data-implied $\hat{\alpha}_S$. (b) Interaction-only cosine at data-matched amplitude ($N = 200$) against the cross-replicate null on the same replicates. The estimator recovers a known operator’s interactions cleanly at the observed signal amplitude.

1.3.3 2.3 The failure is not target selection, input dimension, or κ mis-scaling

Selection bias. The publicly released essential-gene notebooks pick the 200 targets with the most cells after a minimum-cell filter. In essential-gene screens, high cell count correlates with mild fitness defect, so the selection preferentially retains weakly-affected targets. A K562 refit on 200 targets sampled uniformly at random from the 1,740 qualifying targets (all with ≥ 60

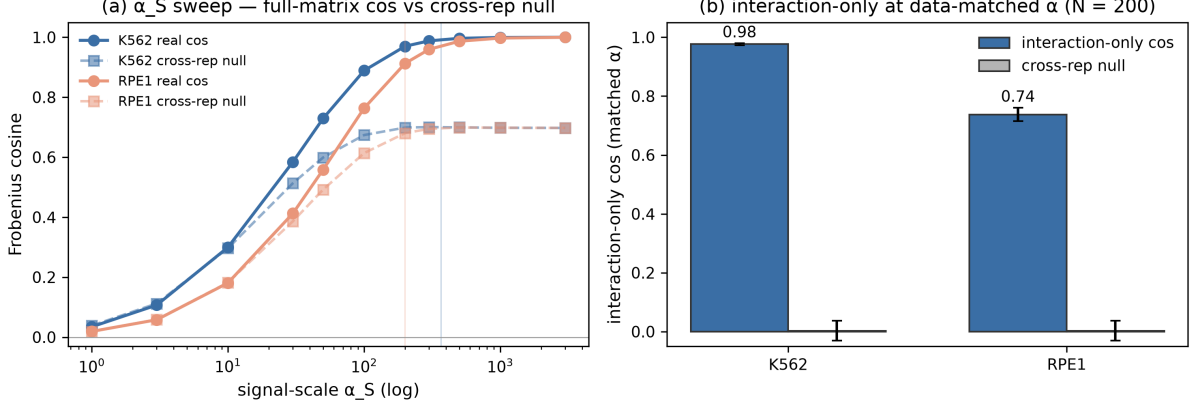


Figure 3: Figure 3

cells) gives real nested-CV $\rho = 1.00$ (5-fold SD 0.009), median picked rank 0, against a matched linear-truth $\rho = 0.43$ (SD 0.011, $N = 15$). The failure survives the random selection.

Design underdetermination. Jost has 25 target directions and $d = 30$ program dimensions; under 5-fold target-grouped CV each outer training set spans only ~ 20 target directions, which is fewer than d . If the encoding failed on Jost only because of this under-determination, the failure should disappear when the coordinate system is truncated to a smaller d where training targets comfortably exceed d . We swept $d \in \{5, 10, 15, 20, 25, 30\}$ on Jost, recomputing α_S at each d so the matched linear-truth control tracks the same signal amplitude at each point.

Table 2. Jost d -sweep under target-grouped nested cross-validation.

d	Real ρ (pooled)	Median picked rank	Matched linear-truth ρ (mean, SD; $N = 15$)	Regime
5	1.01	0	0.32 (0.038)	overdetermined
10	1.00	0	0.85 (0.067)	overdetermined
15	0.99	2	0.46 (0.031)	overdetermined
20	1.00	0	0.55 (0.061)	overdetermined
25	1.00	1	0.69 (0.037)	underdetermined
30	1.00	0	0.73 (0.043)	underdetermined

At $d = 5$, where the identifiability regime is comfortably overdetermined, the matched linear-truth ρ is 0.32 while real Jost ρ is 1.01. The matched linear-truth ρ varies non-monotonically with d — from 0.32 at $d = 5$ to 0.85 at $d = 10$ — because α_S is recomputed at each d and different truncations of the coordinate system carry different fractions of the observed response amplitude; but it stays below the real ρ at every d , so the encoding fails independent of the identifiability regime.

κ mis-scaling. The Replogle-essential pipeline uses a residual-space **detection_rate** proxy for κ (Methods), which is a signed distributional-shift statistic rather than a fractional-knockdown estimate. If mis-scaled per-guide κ were the source of the failure, column-normalizing S and U before the fit — which removes all per-guide scale from the fitting problem — should improve held-out ρ . It does not. Under the same target-grouped nested cross-validation, direction-only real ρ is 0.97 (5-fold SD 0.007) on K562, 1.00 (SD 0.002) on RPE1, and 0.99 (SD 0.017) on Jost. The matched linear-truth control on the same column-normalized design reaches 0.43

(SD 0.018) on K562, 0.78 (SD 0.013) on RPE1, and 0.79 (SD 0.030) on Jost, all well below the predict-zero baseline. Column-normalizing raises the matched-linear ρ on RPE1 and Jost — the transform discards real per-guide scale information those screens’ κ estimators carry (Replogle-essential’s residual-space proxy is uninformative in this regard; Jost’s count-based `mean_ratio` is informative) — but κ mis-scaling cannot on its own account for the encoding’s failure on any of the three screens.

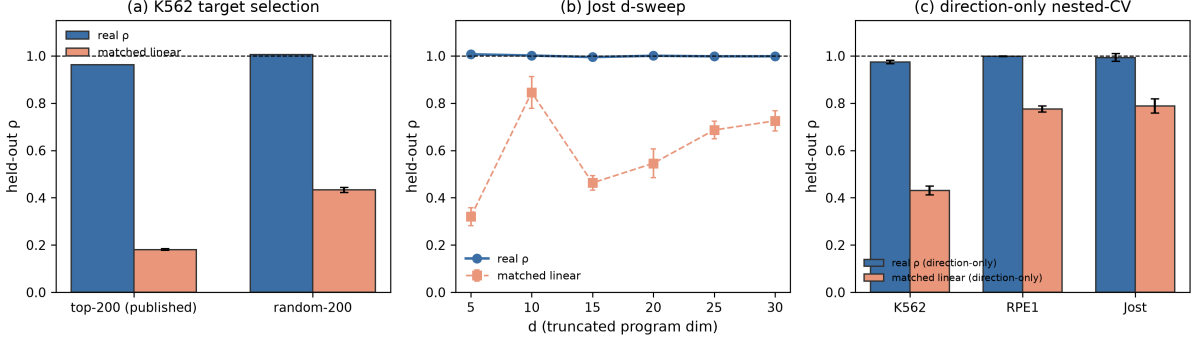


Figure 4: Figure 4

Figure 4. Robustness checks. (a) K562 refit on 200 targets sampled uniformly at random reproduces the failure; the matched linear-truth control drops below 1 at both selections (its α_S is smaller under random-200 because the sampled targets have smaller-amplitude responses). (b) Jost d-sweep under target-grouped nested CV: real ρ stays at the predict-zero baseline across $d \in \{5, 10, 15, 20, 25, 30\}$, while the matched control drops well below 1 in every regime. (c) Direction-only refit under the same nested-CV recipe on all three screens rules out κ mis-scaling.

1.3.4 2.4 The pipeline captures within-target dose interpolation on Jost

Jost 2020 pairs each of its 25 targets with 3–6 sgRNAs at graded activities. Under target-held-out prediction (the setting in §2.1–2.3) a held-out target’s sgRNAs have no siblings in training, and the model is asked to predict a new perturbation direction. Under *guide-level* folds — where every held-out sgRNA has siblings for its own target in training — the model is instead asked to interpolate amplitude along a *known* direction. It predicts held-out U_{test} given S_{test} along that known direction. That interpolation ρ is 0.66 on real Jost, against 0.22 (SD 0.008, $N = 15$) for the matched linear-truth control on the same guide-level folds.

Two things follow. The pipeline does capture within-target amplitude structure when a target’s direction is available: 0.66 is meaningfully below the predict-zero baseline of 1 and well above the null. But the amplitude the pipeline captures is not operator recovery. Under any additive-input linear model, every guide for one target moves the system along the same direction — κ_g rescales columns, $W^\top \delta_g$ is fixed — and the guide-level fold split reduces the prediction task to one that a per-target one-parameter fit could pass. We label the guide-level Jost result “within-target dose interpolation along a known target direction” and keep it separate from the target-held-out claim that governs the main result.

1.3.5 2.5 A widely used linearity threshold is uncalibrated at these signal-to-noise ratios

The `rel_diff` statistic splits guides at the median efficiency, fits an operator per half, and returns the Frobenius-normalized difference on the shared identified subspace. A prespecified threshold `rel_diff` ≤ 0.25 was chosen at preregistration to declare the additive-input model consistent with the data. Real data on Replogle gives `rel_diff` = 1.47 (K562) and 1.57 (RPE1);

on Jost it is 1.26. Every screen fails the threshold. But so does a matched linear ground truth at each screen’s own signal amplitude and noise: 0.33 (SD 0.01) on K562, 0.98 (SD 0.02) on RPE1, and 1.05 (SD 0.03) on Jost ($N = 15$). On RPE1 and Jost even a perfectly linear generator fails the 0.25 threshold; only on K562 does it drop below 1.0. The threshold cannot separate a linear model from a nonlinear one at these signal-to-noise ratios regardless of ground truth. It is miscalibrated. The calibrated comparison — real vs matched linear truth at each screen’s own SNR — corroborates the target-held-out failure on all three: real `rel_diff` exceeds matched linear-truth `rel_diff` by ~ 1.14 (K562), ~ 0.59 (RPE1), and ~ 0.21 (Jost, about 7 linear-truth SDs above the matched-control mean).

We ship a separate correctness fix at the identifiability level: the gate that labels a measurement “full-domain identified” now requires both the input rank and the retained response rank to equal d at the preregistered tolerance. Jost’s input rank is 24 at that tolerance, so the gate now correctly labels the Jost fit as partially identified regardless of what noise does to the retained response rank.

1.3.6 2.6 A footprint encoding does not generally rescue prediction

The additive-input encoding projects a target-direction one-hot through the program basis: $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \delta_g$. If $\sim 99\%$ of the target direction lies outside the retained subspace, the encoded input carries little response-relevant signal. A natural alternative — preregistered before running (Amendment 2026-09-27) — replaces the one-hot with the target’s control-cell co-expression *footprint*: $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \Sigma_{\text{ctrl}} \delta_g$, where Σ_{ctrl} is the gene–gene covariance of non-targeting-control cells only, in the same feature space that fits $\mathbf{W}$. $\Sigma_{\text{ctrl}} \cdot \delta_g$ is the row of the control-cell covariance matrix at the target gene — the target’s neighbourhood of co-expressed genes rather than the target alone.

We ran the footprint encoding on all four fits under exactly the target-grouped nested-CV recipe used for Table 1. The preregistered success criterion was footprint $\rho \leq 0.90$ AND below the 2.5th percentile of a shuffled-footprint null (≥ 100 permutations of the target→footprint mapping). Results are per screen (Table 3).

Table 3. Footprint-encoding held-out ρ vs current encoding and preregistered nulls.

Screen	Current-encoding ρ (Table 1)	Footprint ρ	Shuffled-footprint null 2.5th percentile	Verdict
K562 top-200	0.96	0.81	0.97	success
K562 random-200	1.00	1.00	0.99	failure
RPE1	1.00	1.00	0.98	failure
Jost 2020	1.00	1.00	0.98	failure

Under the preregistered rule, success on at least two screens would have promoted projection failure from an open question to a result. No screen passed cleanly. On K562 the outcome depended on target selection: the footprint encoding beat its shuffled null on the top-200 selection ($\rho = 0.81$ against a 2.5th percentile of 0.97) but not on 200 randomly sampled targets. RPE1 and Jost failed. The K562 top-200 result is selection-dependent, and we did not test why. The preregistered outcome is mixed, and projection failure remains open.

Direction-only nested-CV under the footprint encoding was also computed (column-normalize $\mathbf{S}$ and footprint $\mathbf{U}$ before the nested-CV recipe); it does not change the per-screen verdict.

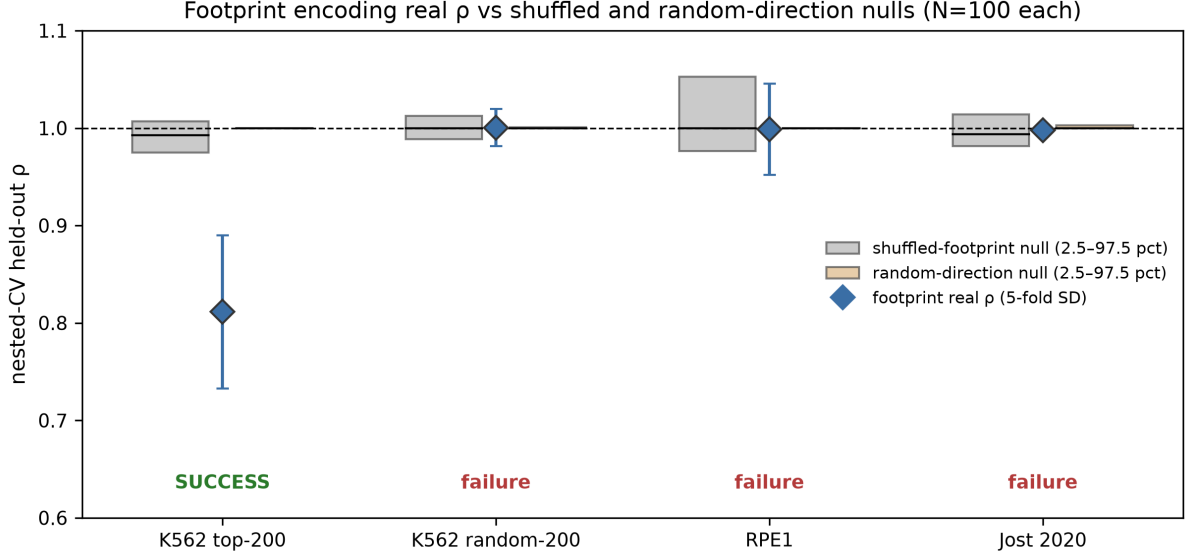


Figure 5: Figure 5

Figure 5. Footprint-encoding nested-CV ρ on the four fits (blue diamonds, with 5-fold SD) against the two preregistered nulls, each drawn as the 2.5–97.5 percentile band with the median as a horizontal bar: shuffled-footprint null ($N = 100$ permutations, grey) and random-direction null ($N = 100$ draws with matched column norms, tan). Success labels (“SUCCESS”/“failure”) reflect the preregistered rule (footprint $\rho \leq 0.90$ AND below the shuffled-null 2.5th percentile). Only K562 top-200 satisfies both conditions.

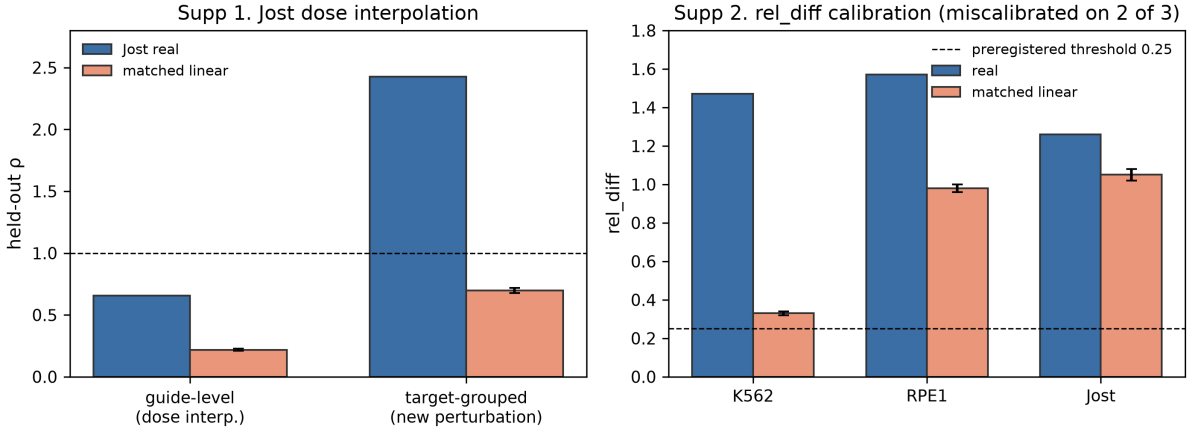


Figure 6: Supplementary Figures

Supplementary Figure S1. (left) Jost dose interpolation: guide-level held-out ρ (siblings in training) is 0.66 for real vs 0.22 for matched linear-truth (SD 0.008); target-grouped held-out ρ is at the predict-zero baseline for both, showing the guide-level ρ is dose interpolation rather than operator recovery. (right) rel_diff calibration on the three screens: the preregistered $\text{rel_diff} \leq 0.25$ threshold is unreachable at every tested SNR even under a matched linear ground truth (RPE1 and Jost), so the threshold cannot separate a linear from a nonlinear model at these signal-to-noise ratios.

1.4 3. Discussion

1.4.1 3.1 What the result says

Under the additive-input encoding $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \delta_g$ and truncated-SVD fitting on three current Perturb-seq screens, target-held-out prediction of a new perturbation does not beat the predict-zero baseline while a matched-SNR linear-truth positive control on the same $\mathbf{U}$, κ , and per-entry noise reaches ρ well below 1 on every screen. Fitted spectra or edges from this encoding should not be read as quantitatively estimated operators on the tested screens; the result does not rule out recovery under an alternative input encoding, an alternative model class, a lower-dimensional target, a stronger structural prior, or a different assay design.

1.4.2 3.2 What is not yet decided

Two candidate mechanisms remain consistent with the data. The footprint-encoding test in §2.6 was preregistered as the direct check on the first; its outcome was mixed: it helped on one K562 target selection and nowhere else, so projection failure is neither supported in general nor ruled out.

Projection failure. The additive-input encoding pushes a target-direction one-hot through a $d = 30$ control-derived program basis. The projected input $\mathbf{W}^\top \delta_g$ has median column norm ~ 0.09 on the essential-gene screens — about 99% of the target direction lies outside the retained subspace. If the target’s response-relevant direction is not among the top d control-derived programs, the encoded input carries little response-relevant signal and the fitting task becomes ill-posed on target-held-out folds. The footprint encoding (§2.6) replaces the one-hot with the target’s control-cell co-expression neighbourhood. It rescues K562 top-200 (real $\rho = 0.81$ vs shuffled-null 2.5th percentile 0.97) but not K562 random-200, RPE1, or Jost.

Dose non-linearity. The linear settled-state assumption $\mathbf{S} = -\mathbf{J}^{-1}\mathbf{U}$ may not describe the assay’s actual dynamics if the response saturates or thresholds in dose. Under any additive-linear model, all sgRNAs for one target move the system along the same direction — κ rescales columns, the direction is fixed. Within-target guide-replicate direction concordance on Jost, compared against a between-target null and a within-guide split-half noise ceiling from the same cell-level bootstraps, tests this directly: strong concordance well above the between-target null and near the noise ceiling is evidence *for* the linear-direction assumption; weak concordance is evidence against it. This diagnostic was not run for this paper and would apply specifically to Jost’s replicate design.

A recent benchmark found that a simple linear model predicts unseen single-gene perturbations in Replogle-type screens about as well as current deep-learning models [10]. Their encoding is learned from training responses; ours was a fixed transform of control-cell covariance. A fixed covariance-based encoding did not generally rescue target-held-out prediction here, so the question their result raises remains open for this framework: whether a learned encoding, fit within training folds only, would. Neither projection failure nor dose nonlinearity is ruled out by what we ran.

Scope. The perturb-seq-era methods cited in the Introduction that operate in *gene* space (rather than in a low-dimensional program space) — for instance, DAG-, instrumental-variable-, and linear-cyclic-latent approaches restricted to a subset of targeted genes [6–9] — avoid the projection step tested here. The target-held-out failure of the additive-input encoding at $d = 30$ does not bear directly on those methods; it is a limitation of scope, not a criticism of a different design choice.

1.4.3 3.3 Limitations

The result concerns three screens, one encoding class, and one estimator family. Structural alternatives — a diagonal-only, block-modular, symmetric, or GRN-constrained operator; a much smaller d ; a footprint or Σ_{ctrl} -weighted encoding; a nonlinear dose-response family; a Perturb-seq design with true replicate targets and calibrated dose titration — were not tested and could behave differently. The identifiability calculation itself, a rank statement about $\text{range}(\mathbf{S})$, is correct as a matter of linear algebra and unaffected by the encoding failure.

Two per-screen caveats. Replogle-essential libraries retain one sgRNA per target after the released aggregation step, so within-target replicate diagnostics do not apply to them. Jost has 25 targets, small enough that a target-held-out CV places most of the design outside the training set and the fit sits close to under-determined at $d = 30$; the d -sweep controls for this at $d = 5$ but does not turn Jost into a general genome-scale test. Cross-screen quantitative comparison of specific ρ values requires care because per-cell noise structure differs across cell lines.

The additive-input model itself is a modelling choice. CRISPRi is closer to a clamp on target transcript than to an additive forcing term. Under a program-space clamp with orthonormal $\mathbf{W}$ and a rank-deficient $\mathbf{J}_{\text{gene}} = \mathbf{W} \cdot \mathbf{J}_{\text{prog}} \cdot \mathbf{W}^\top$, the intervention model is exactly under-identified from projected observations. This is a valid algebraic result under those assumptions rather than a general impossibility; alternative encoder/decoder conventions or explicit gene-space dynamics can change the conclusion. CRISPRi is often promoter-specific, and untargeted alternative promoters can compensate [11], so a knockdown efficiency measured for one transcript may misstate the effective input; the additive-input encoding assumes one input direction per target.

We report every real held-out ρ with its across-outer-fold standard deviation and every simulated positive-control ρ with its cross-replicate standard deviation. Interpretation carries the same reservation: held-out ρ on a real Perturb-seq screen is the quantity that a matched-SNR linear-truth positive control makes calibrated.

1.5 4. Methods

1.5.1 4.1 Framework

For expression $\mathbf{E} \in \mathbb{R}^{(\mathbf{n} \times \mathbf{G})}$ and a control-derived program basis $\mathbf{W} \in \mathbb{R}^{(\mathbf{G} \times \mathbf{d})}$ with orthonormal columns, program coordinates are $\mathbf{z} = \mathbf{W}^\top \mathbf{e}$. For a guide targeting g at efficiency $\kappa_g \in (0, 1]$, the perturbation input is $\mathbf{u}_g = -\kappa_g \mathbf{W}^\top \delta_g$; stacking m guides gives $\mathbf{U} \in \mathbb{R}^{(\mathbf{d} \times \mathbf{m})}$. Under the additive-input, steady-state assumption $\mathbf{S} = -\mathbf{J}^{-1}\mathbf{U}$, so $\mathbf{J} \cdot \mathbf{S} = -\mathbf{U}$. Right-multiplying by the pseudoinverse identifies the operator’s action on $\mathbf{X} = \text{range}(\mathbf{S})$:

$$\mathbf{J} \cdot \mathbf{P}_{\mathbf{X}} = -\mathbf{U} \cdot \mathbf{S}^+$$

$\mathbf{J}$ is not identified outside $\mathbf{X}$. anchor-op returns this action and blocks access to the full $\mathbf{J}$ unless both the actuated input subspace $\text{range}(\mathbf{U})$ and the retained response subspace $\text{range}(\mathbf{S})$ have rank d at the preregistered tolerance (§4.2).

1.5.2 4.2 Regularization and the identifiability gate

$\mathbf{S}^+$ is computed by truncated SVD, with the full regularization path retained. A singular direction counts as identified only if $\sigma_i > \text{rank_tol} \cdot \sigma_{\text{max}}(\mathbf{S})$. The default $\text{rank_tol} = 1 \times 10^{-2}$ was preregistered to prevent eps-based numerical rank from silently accepting below-noise directions on collinear guide libraries. The identifiability gate now labels a fit as full-domain-identified only when both $\text{input_rank}(\mathbf{U})$ and the effective retained response rank equal

d at the same tolerance; noise-inflated retained rank on rank-deficient U was previously accepted and is not. Regression tests cover both the noise-inflated failure mode and the heavy-Tikhonov collapse.

1.5.3 4.3 Efficiency estimation

Three estimators with an auto-router. `mean_ratio` ($\kappa = 1 - \text{mean_pert} / \text{mean_ctrl}$) is asymptotically unbiased under Poisson and consistent under independent zero-inflation. `poisson_mle` matches under pure Poisson. `detection_rate` ($\Pr[X_{\text{ctrl}} > 0] - \Pr[X_{\text{pert}} > 0]$) is not a κ estimator on count data but is a valid signed distributional-shift statistic on pre-scaled residual data. The "auto" router chooses `detection_rate` when $\geq 2\%$ of entries are negative (the pre-scaled-residual signature), else `mean_ratio`. On Replogle-essential's pre-scaled residual matrix the router chooses `detection_rate`; on Jost 2020's UMI counts it chooses `mean_ratio`. Where the manuscript reports a per-screen κ distribution, Replogle values are the proxy score and Jost values are the count-based estimate; they are not commensurable on a common fractional-knockdown axis and are not compared as such.

1.5.4 4.4 Per-entry noise anchors

Within-guide cell-level split-half bootstraps on each screen: for each target with ≥ 20 cells, split cells into equal halves, compute half- Δz vectors d_1, d_2 in that screen's own PCA basis fit on non-targeting controls, and take the median per-entry standard deviation $\|d_1 - d_2\|_F / (2\sqrt{d})$ across targets. This gives $\sigma = 0.24$ (K562), 0.35 (RPE1), and 0.066 (Jost, per sgRNA). The three anchors are used in every matched-SNR positive control (§4.6).

1.5.5 4.5 Target-held-out nested cross-validation

Outer folds: 5 target-grouped folds. All sgRNAs of a target are held out together, so no sibling sgRNA leaks between train and test. Inner folds within each outer training set: 3 target-grouped folds by the same rule. The TSVD rank is chosen per outer fold from the grid $\{0, 1, 2, 3, 5, 8, 12, 16, 20, 25, 30\}$ by minimizing pooled inner-MSE of $A \cdot S_{\text{val}} + U_{\text{val}}$. $r = 0$ returns $A \equiv 0$ and gives outer-fold $\rho = 1$ exactly, which lets nested CV select "shrink to zero" whenever the fitted model does worse than predict-zero on inner validation. We refit at the selected rank on the full outer training set and report pooled ρ across outer folds with the across-outer-fold standard deviation.

The Jost d -sweep in §2.3 truncates S and U to the first $d' \in \{5, 10, 15, 20, 25, 30\}$ rows before running the same recipe. The K562 random-200 refit in §2.3 rebuilds the entire pipeline (variance-based highly-variable-gene selection with the sampled target genes force-kept in the feature matrix, PCA on non-targeting controls at $d = 30$, per-guide response construction with the proxy-efficiency path used in the released essential-gene notebooks) on 200 targets sampled uniformly at random from the 1,740 targets with ≥ 60 cells.

1.5.6 4.6 Matched-SNR linear-truth positive control

For each screen we compute the observed median column norm of S . For each of 15 independent simulations, we draw $J_{\text{ref}} = G - 1.5 \cdot I$ with G entrywise $N(0, 1/d)$ and set $J = J_{\text{ref}} / \alpha_S$, choosing α_S so the median column norm of the noise-free $S_{\text{true}} = -J^{-1}U$ matches the observed median at each screen's own U . We add per-entry noise at the screen's bootstrapped σ (§4.4) and run the same nested-CV recipe (§4.5) on this synthetic (S_{obs}, U) pair. We report the mean and standard deviation of held-out ρ across the 15 simulations.

The interaction-only cosine reported in §2.2 uses the same synthetic ground-truth ensemble at matched α_S and reports Frobenius cosine after subtracting the diagonal from both A and J ,

with a cross-replicate null as its baseline. The cross-replicate null pairs each fitted $\mathbf{A}_r$ with an independently drawn ground truth $\mathbf{J}_{\{r'\}}$ (shift-1 pairing in the replicate index; $N = 200$ replicates), so every element of the pipeline is preserved and only the ground truth we score against is randomized. The signal-scale sweep in §2.2 varies α_S across a log grid and reports the cos-versus- α_S curve and the α_S implied by the observed median column norm.

1.5.7 4.7 Direction-only refit

Column-normalize $\mathbf{S}$ and $\mathbf{U}$ — divide each column by its own Frobenius norm — then run the same nested-CV recipe (§4.5). This removes all per-guide scale from the fitting problem. For the matched-SNR linear-truth control we generate $\mathbf{S}_{\text{obs}} = \mathbf{S}_{\text{true}} + \sigma \cdot \varepsilon$ at each screen's own α_S and σ on the same $\mathbf{U}$, column-normalize both $\mathbf{S}_{\text{obs}}$ and $\mathbf{U}$ in the same way, and run the same recipe. Both are reported in §2.3.

1.5.8 4.8 Preregistered vs post hoc

Preregistered: $\text{rank_tol} = 1 \times 10^{-2}$, TSVD regularization, $d = 30$ PCA basis fit on non-targeting controls, the raw bin-split $\text{rel_diff} \leq 0.25$ threshold. Post hoc: per-screen σ anchors (§4.4, replacing an earlier common σ); the α_S signal-scale sweep and matched- α positive control (§4.6); the target-grouped nested-CV recipe and the rank-0 baseline (§4.5); the interaction-only cosine analysis (§4.6); the direction-only refit (§4.7); the K562 random-200 refit and the Jost d-sweep (both executed and reported in §2.3); the identifiability gate now requiring $\text{rank}(\mathbf{U}) = d$ as well as retained $\text{rank}(\mathbf{S}) = d$, a bug fix affecting the identifiability label on rank-deficient inputs with no numerical-recovery changes. A preregistration amendment is shipped with the release.

1.5.9 4.9 Data availability

Replogle et al. 2022: [gwps.wi.mit.edu / Figshare Plus deposit 20029387](https://gwps.wi.mit.edu/) (K562 essential sampled at day 6; RPE1 essential sampled at day 7) [1]. Jost et al. 2020: GEO GSE132080 [2].

1.5.10 4.10 Code availability

anchor-op is available at <https://github.com/manarai/anchor-op> under the MIT license. A pinned conda `environment.yml` (Python 3.11), a 59-test pytest suite, per-figure regeneration scripts, and every JSON that a numeric claim traces to are shipped with the release. Every numeric claim in this manuscript that is not derived in the text has an entry in `results/recheck/number_provenance.csv` giving the value stored in a JSON in `results/recheck/`, the reproduction script that produced it, and a note on rounding where applicable. Round-form numbers cited in the abstract and Results are drawn from the un-rounded values in that table.

1.6 Author contributions

Dalton Kutzen: Methodology, Formal analysis, Investigation, Writing — original draft, Writing — review & editing.

Sam Green: Validation, Visualization, Writing — review & editing.

Tommy W. Terroatea: Conceptualization, Supervision, Project administration, Funding acquisition, Writing — review & editing.

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1.8 Competing interests

The authors declare no competing interests.

1.9 References

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